

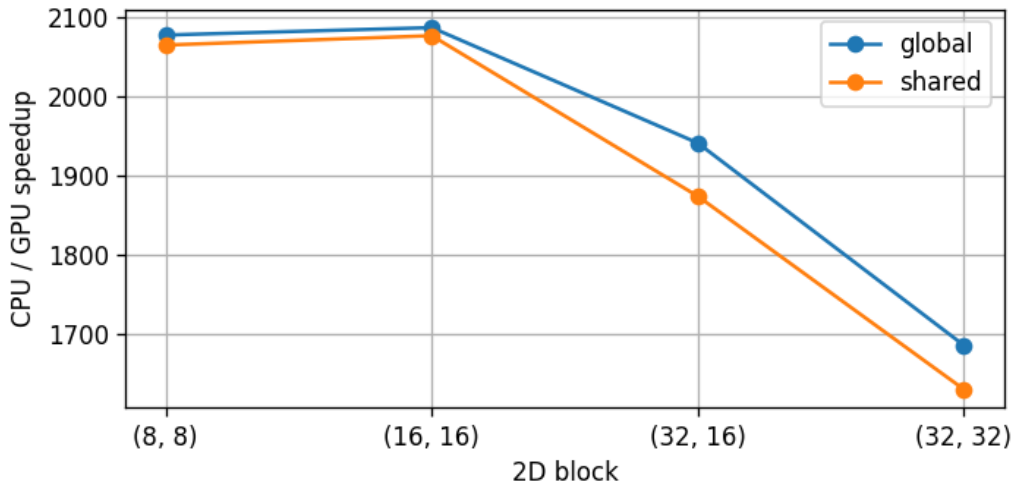
Lab 5: Gaussian Blur

Nguyễn Thanh Hùng – 2540056

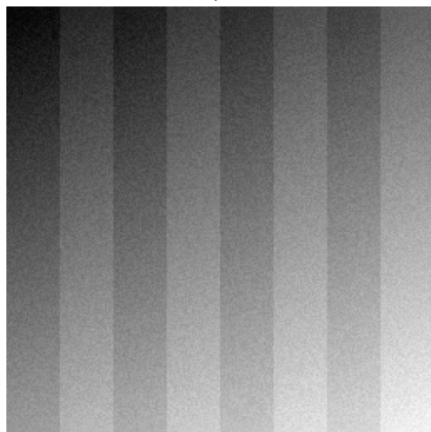
A 7×7 Gaussian filter with coefficient sum 1003 is applied to grayscale pixels. The global version reads the filter from device memory; the shared version cooperatively loads its 49 coefficients and synchronizes before convolution. The three-pixel border is preserved. A CPU reference validates both versions. The graph compares speedup for different 2D blocks. Shared memory has loading and synchronization costs, so its benefit must be measured.

Kernel timing uses `time.time()` after JIT warm-up and CUDA synchronization, averaged over three GPU runs. Device allocation and host/device copies are excluded. CPU measurements are single runs. Speedup is relative to the stated CPU implementation. Actual measurements appear after running the notebook on Colab; no timings are assumed here.

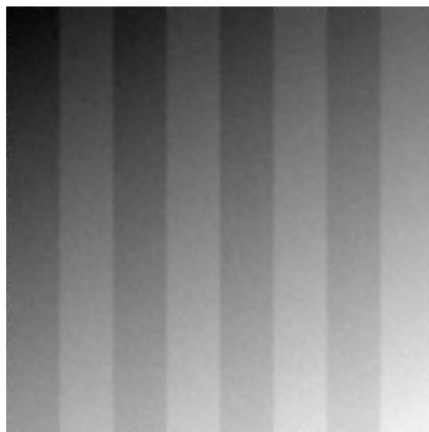
block	mode	cpu_ms	gpu_ms	speedup	max_error
(8, 8)	global	404.677	0.194709	2078.37	0
(8, 8)	shared	404.677	0.195901	2065.72	0
(16, 16)	global	404.677	0.193834	2087.75	0
(16, 16)	shared	404.677	0.194788	2077.52	0
(32, 16)	global	404.677	0.208378	1942.03	0
(32, 16)	shared	404.677	0.215848	1874.82	0
(32, 32)	global	404.677	0.239929	1686.65	0
(32, 32)	shared	404.677	0.248114	1631.01	0



Input



Global



Shared filter

